25th International Conference of Young Scientists – ICYS 2018 Belgrade

GENERAL LONGBOARD SCIENCE
HOW TO BUILD YOUR OWN LONGBOARD

Authors: Jelle Feitsma, Sam Voncken - *jellefeitsma@gmail.com ; sam.voncken@hotmail.nl*

Supervisor: Leonie Titulaer, PhD; Porta Mosana College Maastricht, the Netherlands

Introduction

In this profile project, we combined design, research and construction in one assignment: we constructed and tested a self-designed cruiser-type longboard. Designing and building a longboard involved researching a number of important aspects related to physics and chemistry and included some biological and societal considerations: we aimed to carry-out this project in a sustainable manner. For this reason, this project is entitled: "*General Longboard Science*".

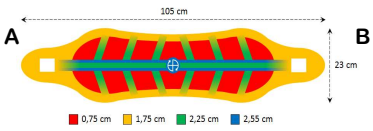
Experimental Setup

We have designed and produced all important parts of the longboard ourselves: this included the deck, the trucks and the wheels. We also had to construct our own tools and molds to accomplish this profile project. For each part of the longboard, a design-cycle was followed: a) global analysis and description of the design, b) definition of a program of requirements (properties and dimension), c) devising a practical approach and formulating the design, d) realize the design and e) testing and evaluating the final product. This systematic approach enabled us to determine how the parts of the longboard could be made and which preparations and actions were needed. Each final product was compared to the original program of requirements.

Conclusion

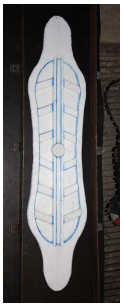
What we have learned from this project is that it is absolutely necessary to thoroughly devise and elaborate all steps in advance. We have benefited greatly from the availability of a 3D-printer at school. Looking back, avoiding bubbles and releasing the deck from its mold provided the biggest challenges. We have absolutely enjoyed making the trucks: thorough preparation of sandcasting paid-off in a near-perfect set of trucks. The outcome of the X-ray analyses (truck and wheels) was a highlight of this project. A very instructive aspect of this project has been clear and motivated communication with professionals. The enthusiastic participation of these experts has been pivotal and very much appreciated by us.

the Deck







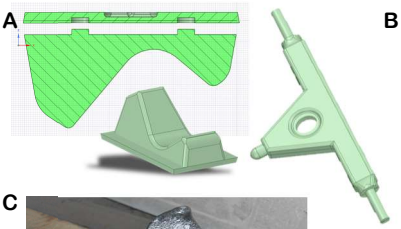


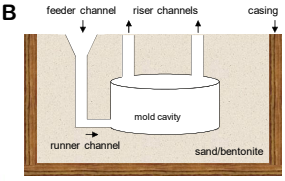




The deck - A) deck design: (*top*) colour codes correspond to differences in material thickness; stability and strength were achieved in length and in double concave/camber design. A spine/rib structure along the longitudinal axis of the deck was based on the nervature that supports a banana leave (*bottom left*). The deck was made from recycled, shredded High-Density PolyEthylene (HDPE; courtesy Stiphout Plastics, NL; *bottom right*). B) a styrofoam real-size deck prototype (*left*) was used to construct a deck-mold in concrete (*right*). C) molten HDPE was pressure-molded into a self-constructed concrete/steel mold under ±1.5-2 ton pressure and left to cool. Excess plastic was trimmed; the deck was finished by sanding and varnishing.

the Trucks





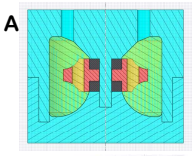


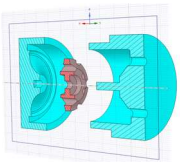




The trucks - A) truck design: (*top*) the baseplate (*left*) and hanger plastic prototypes (*right*) were designed using *DesignSpark Mechanical* software and generated using a 3D-printer. These prototypes were used for sand-casting the final aluminum parts which connect the wheels to the deck. B) (*top*) schematic representation of sand-casting; (*middle*) plastic prototypes were pressed into a home-made sand-casting mold, in which the feeder and riser channels are clearly visible; (*bottom*) a self-constructed crucible and melting oven were used to melt and pour the aluminum alloy (courtesy MGG, NL) it into the sand-casting mold. C) Upon setting of the aluminum alloy cast, excess metal was removed and the trucks were finished by grinding and filing.

The Wheels & Bushings

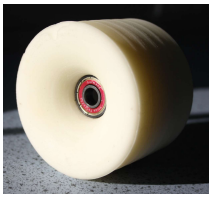






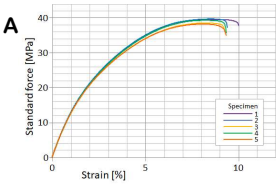




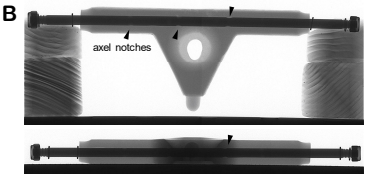


The wheels and bushings - A) wheel design: longitudinal section (*top left*) *and* cross-sectional worked-open virtual 3D-print models in which the wheel core is depicted in grey (*bottom left*); bushing (*top middle*) and wheel (*top right*) molds were designed using *DesignSpark Mechanical* software and 3D-printed using biodegradable polylactic acid. The ABS (acrylonitril-butadieen-styreen) hard plastic wheel cores (*bottom middle*) were also designed and generated by 3D-printing. B) The polyurethane (courtesy Romar-Voss, NL) wheels were cast with the wheel-cores inside (*see also: figure A, top right*) that held the ball-bearings, which in turn ensure friction-free wheel rotation (the ball-bearings were generously provided by Skateshop Titus, Aachen, GE).

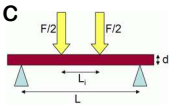
Scientific analysis







Scientific analyses. The quality of the end products was scientifically tested: A) (*top left*) tensile strength diagrams of HDPE plastic strips (*top right*) were generated using specialized equipment (courtesy CHILL, Chemelot, NL). B) the presence of artefacts (air bubbles) in the hangers was excluded by Röntgen analysis (courtesy MGG, NL). C) four-point flexural strength was estimated based on the dimensions of the board and the measured tensile strength. D) Endproduct: a self-made cruiser-type longboard.



$$\sigma_f = \frac{3 \cdot F \cdot (L - Li)}{2 \cdot b \cdot d2}$$

